Pick-and-place with Franka and Heal

Gaurav Kumar Srivastava | Roll number: 23110113
Lab 3 - ME 639 Introduction to Robotics
Professor Madhu Vadali | AY 2026-27

1. Introduction and goals

This laboratory assignment develops a complete pick-and-place operation in MuJoCo: locate a cube on a table, grasp it, carry it over a tray and release it successfully. The task was implemented on the seven-joint Franka and six-joint Heal arms. Mink provides the baseline inverse kinematics (IK) controller; custom damped least squares (DLS) and quadratic programming (QP) controllers are compared on the same task.

The main challenge is to turn a reachable end-effector pose into a reliable physical action. The gripper must align with a randomly oriented cube, maintain the grasp during motion and clear the tray rim before descending. Workspace analysis therefore guides the table dimensions and height, while the simulations test whether the complete motion succeeds.

2. Methods

2.1 Environment and task execution

Each scene contains the supplied arm, a level table, a free 40 mm cube of mass 50 g, and an open tray. Franka retains its parallel-jaw gripper. Heal uses the MuJoCo Menagerie Robotiq 2F-85 attachment because meshes for the supplied custom gripper were missing. Physics advances at 500 Hz, with IK and actuator commands updated at 100 Hz.

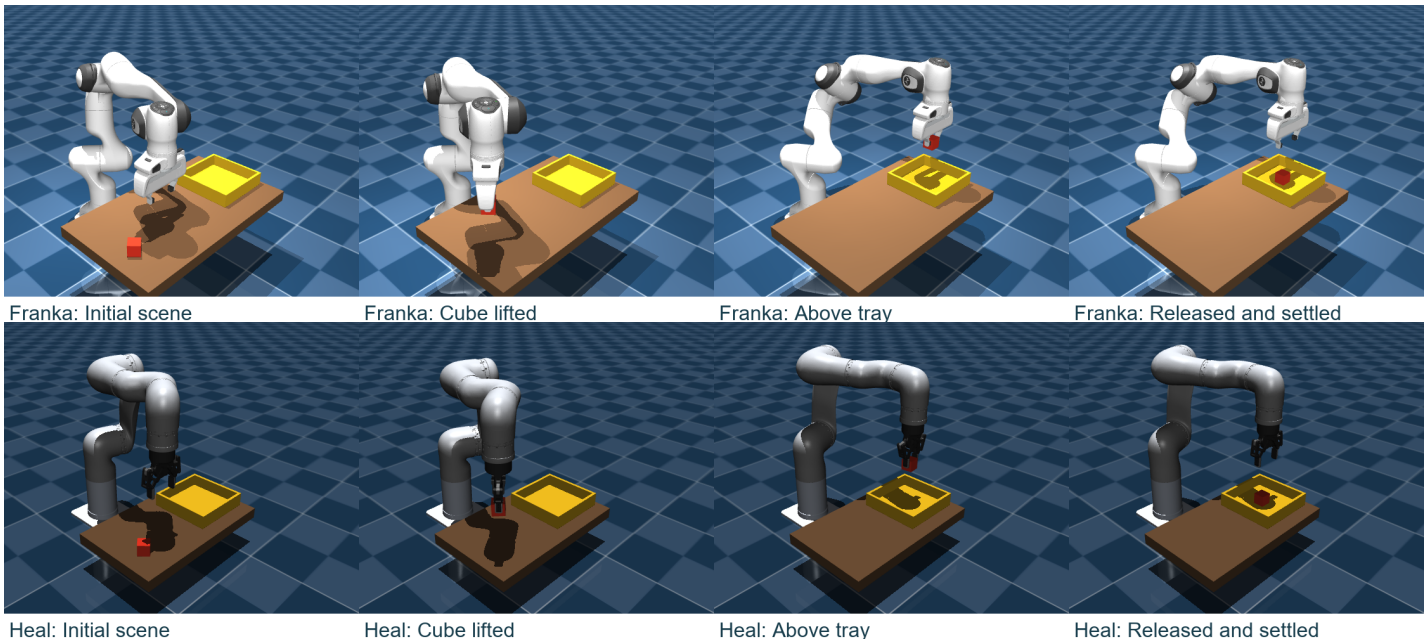


Figure 1. Successful DLS episodes on the workspace-sized tables (seed 3100). Each row follows one robot from the initial scene through lifting and transfer to a released, settled cube.

The controller first settles the scene, approaches vertically, descends and closes the fingers. It then lifts the cube, transfers above the tray, lowers for release, opens the fingers and retreats. A lift-height check prevents a failed grasp from being treated as a successful transfer. Figure 1 shows both arms completing this sequence through actuator-driven contact dynamics.

2. Methods (continued)

2.2 Workspace analysis and table design

A position that is reachable in some posture may not allow a downward grasp at the required wrist angle. To separate these questions, 8,000 joint configurations were sampled for each arm and mapped by forward kinematics (FK). Downward-facing tool centre point (TCP) poses were then tested across an XY grid at seven table heights, from 0.10 to 0.40 m. A point was accepted as yaw-dexterous only when IK succeeded at all eight tested yaw angles.

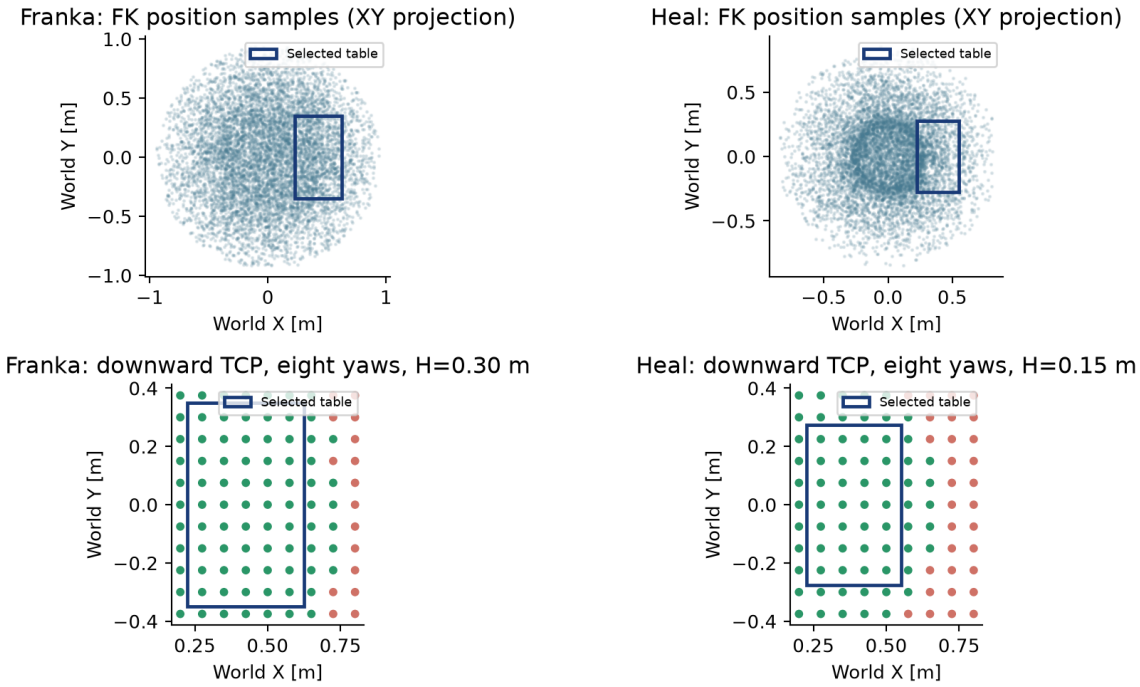


Figure 2. Top: XY projections of the FK position samples. Bottom: downward-grasp feasibility at the selected heights; green points pass all eight yaw tests and red points fail at least one. Blue outlines show the workspace-sized tables. The top plots show positional reach, not collision-free task feasibility.

The table footprint was chosen as the largest packable rectangle found within these sampled feasible regions. The search used a 0.075 m XY grid and rectangle orientations at 15-degree intervals, followed by finer footprint checks. Candidates also had to accommodate the unchanged 0.22 m tray footprint and a separate cube-placement band. Approach and lift poses were checked as well as grasp height, so the selected surface supports the motion rather than just its endpoints at table level.

Table design	Length x width [m]	Height [m]	Centre X, Y [m]	Yaw
Workspace-sized: Franka	0.40 x 0.70	0.30	0.425, 0.000	0 deg
Workspace-sized: Heal	0.325 x 0.55	0.15	0.3875, 0.000	0 deg
Additional placement: Franka	0.50 x 0.60	0.30	0.550, -0.100	0 deg
Additional placement: Heal	0.50 x 0.60	0.15	0.480, -0.070	30 deg

The different heights reflect the arms' different downward-grasp workspaces; a common table was not imposed. As an additional study, height, horizontal position and in-plane rotation were varied for a fixed 0.50 x 0.60 m table. Those layouts are retained for comparison, but table sizing is the main design exercise here. Neither study tilts the tabletop or establishes a global optimum.

2. Methods (continued)

2.3 Randomization and differential IK

Cube coordinates are sampled uniformly within the table-local pick region, and yaw over a full turn. The controller uses the square cube's equivalent face grasps to reduce unnecessary wrist rotation without changing its pose.

At each update, the measured TCP pose gives a six-component error: desired minus current position, followed by the shortest rotation vector from the relative rotation matrix. The Jacobian J maps joint velocity v to Cartesian velocity; the custom methods request a correction of 4e:

$$\mathbf{e} = [(\mathbf{p}_d - \mathbf{p})^T,\ \mathrm{rotvec}(R_d R^T)^T]^T, \qquad J\mathbf{v} \approx 4\mathbf{e}$$

$$\mathrm{DLS}: \quad (J^T J + \lambda I)\mathbf{v} = J^T(4\mathbf{e})$$

$$\mathrm{QP}: \quad \min_{\mathbf{v}} \ \tfrac{1}{2}\|J\mathbf{v} - 4\mathbf{e}\|^2 + \tfrac{\lambda}{2}\|\mathbf{v}\|^2, \quad \mathbf{l} \le \mathbf{v} \le \mathbf{u}$$

Here p and R describe the current TCP pose; subscript d denotes the target. The correction gain is 4 s^{-1} and damping uses lambda = 0.001. DLS solves the linear system without forming an inverse.

Mink uses an SE(3) FrameTask, configuration/velocity limits and quadprog. **DLS** computes a damped velocity, then clips it to the bounds; damping reduces excessive corrections near poorly conditioned configurations. **QP** enforces the bounds during optimization, allowing correction to be redistributed across other joints. Without active bounds, the custom methods agree.

Joint bounds are l = max[-0.6, (q_{min} + epsilon - q)/dt] and u = min[0.6, (q_{max} - epsilon - q)/dt], where dt = 0.01 s and epsilon = 0.0001 rad. Reaching a waypoint requires ten consecutive updates below 3 mm position and 0.04 rad orientation error. Franka uses position actuators with bias feedforward and a 6 rad/s^2 command-slew limit; Heal uses effort tracking with inertia, bias, damping and friction compensation.

These tracking layers convert the common IK velocity output into commands suitable for each arm. The slew limit bounds the change in Franka's requested velocity, not its physical acceleration; measured joint motion is checked separately.

2.4 Avoiding the tray rim

A direct low path risks striking the 0.05 m tray rim. Instead, the TCP lifts vertically to 0.18 m above the table, translates and aligns the jaws over the tray, then lowers to 0.07 m for release. Joint-bounded QP alone does not avoid obstacles: the staged path provides clearance, and actual cube/wall distance is measured during physics.

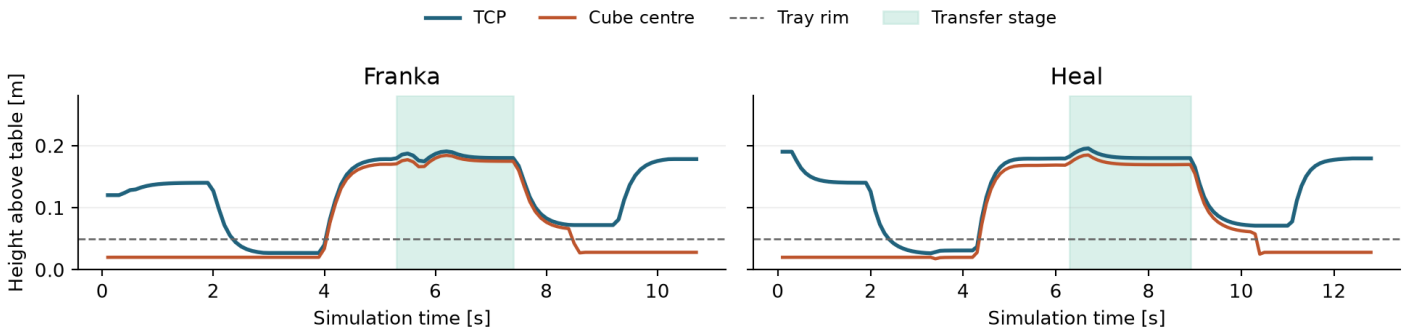


Figure 3. Heights during the Figure 1 episodes. The cube clears the rim during transfer (shaded), then settles as the TCP retreats. Centre height is not surface-to-surface clearance.

3. Results and comparison

Each robot/method combination uses the same 25 seeds within a study. Success requires a released cube within 52 mm and 47 mm of the tray centre along its local axes, at 22-40 mm centre height above the table and below 0.03 m/s linear speed. Constraint checks run every 2 ms; intended grasp and support contacts are allowed.

Workspace-sized tables: main comparison

Robot / IK	Placed / 25	IK time [ms]	IK updates	Episode [s]	Min wall [mm]	Peak speed [rad/s]	J / V / C
Franka / Mink	25	565.3	747.8	10.30	67.10	0.809	0/0/0
Franka / DLS	25	298.9	700.6	9.83	65.70	0.791	0/0/0
Franka / QP	25	366.4	743.9	10.26	67.25	0.795	0/0/0
Heal / Mink	25	747.8	1138.0	14.20	71.45	0.682	0/0/0
Heal / DLS	25	363.1	1076.8	13.59	58.40	0.804	0/0/0
Heal / QP	25	444.6	1133.4	14.15	71.50	0.679	0/0/0

Fixed-size table placement: additional comparison

Robot / IK	Placed / 25	IK time [ms]	IK updates	Episode [s]	Min wall [mm]	Peak speed [rad/s]	J / V / C
Franka / Mink	25	441.7	738.8	10.21	72.61	0.786	0/0/0
Franka / DLS	25	224.1	698.7	9.81	73.52	0.791	0/0/0
Franka / QP	25	287.2	735.0	10.17	72.18	0.771	0/0/0
Heal / Mink	25	601.8	1079.7	13.61	69.77	0.724	0/0/0
Heal / DLS	25	290.8	965.2	12.47	51.31	0.684	0/0/0
Heal / QP	23	392.3	1068.2	13.30	69.14	0.723	0/0/0

IK time, update count and simulated duration are episode means. Updates are controller calls, not QP iterations. Min wall is the group minimum cube/wall distance; peak speed is the largest measured joint speed. J/V/C count episodes with joint-limit, speed (>1.05 rad/s), or moving-arm/environment-contact violations. All successful placements passed these checks. Physical speed can exceed the 0.6 rad/s command bound. Seeds: 3100-3124 (sizing), 2000-2024 (placement).

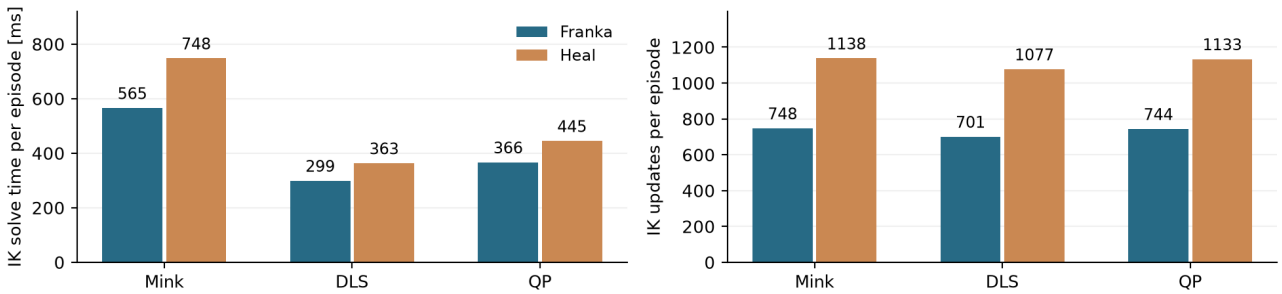


Figure 4. Mean IK solve time and update count on the workspace-sized tables. Solve time measures computation; episode duration includes motion and waiting.

All methods completed 25/25 episodes on both workspace-sized tables. DLS had the lowest IK time: 298.9 ms for Franka and 363.1 ms for Heal. Avoiding constrained-solver overhead is a plausible contributor. QP cost more computation than DLS but less than Mink, while Mink provided a convenient task-and-limit interface.

Heal needed more IK updates and longer simulated episodes than Franka. Arm geometry, table height and actuator tracking all differ, so this is a system-level observation rather than an isolated effect of joint count. In the additional placement study, Heal QP completed 23/25 episodes; the two failures are discussed next. The studies use different pick regions and seeds, so their success counts do not by themselves establish that one table design is superior.

4. Failure analysis and mitigations

Early Franka trials showed abrupt commands, speed peaks and failed lifts. This motivated the command-slew limit and the explicit lift check before transfer. The final comparisons recorded no measured-speed-threshold violations. The earlier failure remains useful because it shows why solving IK alone is insufficient: actuator tracking and contact behavior also determine whether a grasp succeeds.

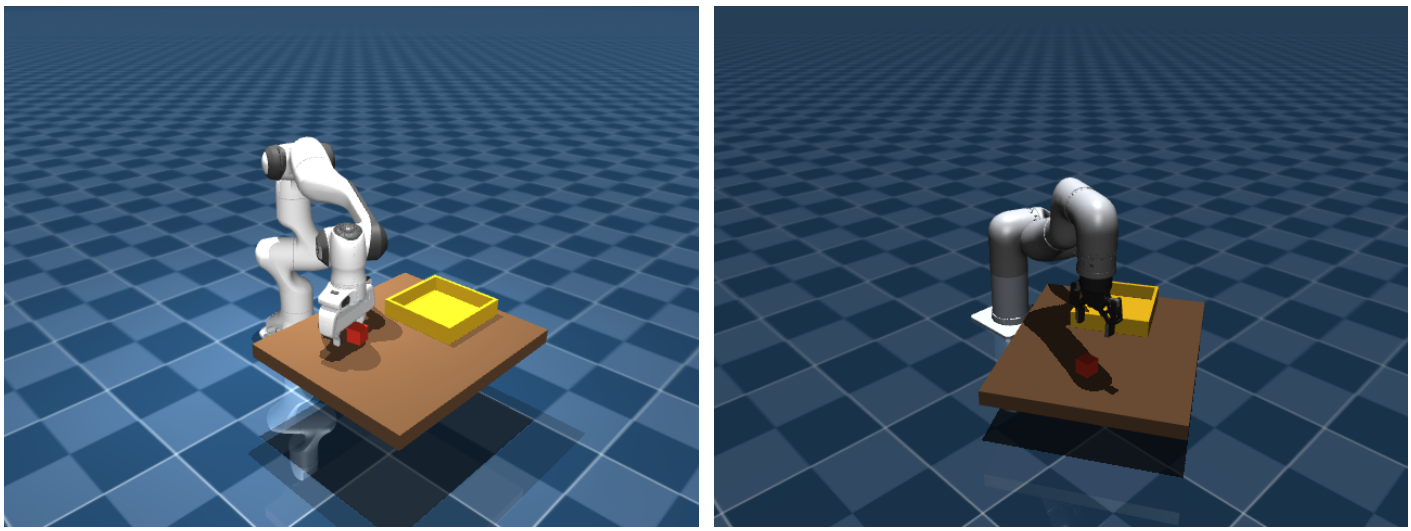


Figure 5. Left: earlier Franka Mink failure before command-slew limiting (seed 1005). Right: Heal QP pre-grasp timeout on the additional fixed-size layout (seed 2004); the cube has not been picked up.

The two remaining Heal QP failures occurred at pre-grasp, not during transport or release. At the 12 s timeout, position errors were 37.8 and 39.1 mm for seeds 2004 and 2018, while orientation errors were only 0.0067 and 0.0069 rad. Thus the approach failed mainly to converge in position despite satisfactory orientation alignment. The logs do not establish whether joint bounds, the chosen IK branch or physical tracking caused the residual.

Issue	Implemented response or next step
Abrupt Franka commands / failed lift	Implemented: command-slew limiting and cube-height verification before transfer.
Tray-rim collision risk	Implemented: vertical lift, high transfer, jaw alignment and centred lowering; monitor cube/wall distance.
Heal QP pre-grasp timeout	Proposed: examine active joint bounds and position error over time, then test an intermediate approach waypoint before changing the timeout.

5. Discussion and conclusions

The task was completed on both arms with Mink, DLS and QP. Workspace-based table selection provided feasible grasp, approach and lift regions, while the staged transfer addressed the tray-rim hazard. Across the main evaluation, all 150 episodes placed the cube successfully and passed the recorded constraint checks. DLS was computationally cheapest in this implementation; QP offered explicit bound handling, but that alone did not guarantee convergence in every additional-layout trial.

The main lesson is that IK, actuator tracking and contact-aware sequencing must be designed together. A low pose error is useful only if the fingers establish a grasp and the carried object follows a clear path. Further work should focus on the Heal pre-grasp residual and broader collision checking rather than simply increasing trial counts. The results apply to these MuJoCo models and sampled poses; hardware behavior was not tested.

References: Course assignment, Tutorial 4: Pick and Place of Objects; Mink; MuJoCo documentation; MuJoCo Menagerie (Robotiq 2F-85 model).